

G02 / ASSESSMENT

G02 GATE B

Standalone Zero-to-Job-Ready Data Engineer 2026

Study, practise, check and explain.

A complete learning route

Start with the lesson video. Read the worked explanation beside it, pause to reproduce the example, then solve the supplied task. The books contain your study notes; keep your own work and results as evidence.

Use the protected answers after saving your first attempt. A changed retry shows what you can now do independently. There is no forced timer during learning.

One course. M00-M19. 264 lessons and 50 project phases.

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Read this first

YOUR STANDALONE GATE BOOK

Different SQL + PostgreSQL reassessment - open only after review and repair

Assessment rule: Save the current attempt before opening its review. Fresh Gate B stays closed until explicitly assigned after repair.

Open only after review and repair

This is a fresh full reassessment after M02-M03. Open it only after Gate A has been saved, Review A has identified material weaknesses, targeted repair is complete, and you are ready for a different assessment. Keep this review and any solution material closed during the attempt.

Load the supplied G02_GATE_B_SETUP.sql into your personal standalone_de PostgreSQL database. The setup script creates fictional course.g02b_* tables only. Record the database/role you actually used and preserve your first .sql attempt even if it is wrong.

Business contract

g02b_orders is one row per order. g02b_items is one row per order item line. An order may have no item lines. customer_id may be missing on an order. is_paid may be TRUE, FALSE or NULL, and NULL means unknown. shipping_fee belongs to the order header and must be counted once per order.

Eight independent deliverables

1. Source controls. Prove source row counts, item gross and original header-fee total directly from the base tables before building a large query.
2. Order-grain result. Create a CTE-based order result containing every order, item gross, shipping fee, order gross, payment group and region/Unknown bucket. Preserve empty orders and missing customer identity.
3. Reconciliation. Prove order-row uniqueness and reconcile item gross + original fees = final order gross. Explain why summing fees after a raw item join can be wrong.
4. Grouped reasoning. Return region-level order count, gross and mean order gross; then keep regions whose gross is at least 90. State both the input and output grain. Compare the overall mean order gross with an unweighted average of region means and explain any difference.
5. Windows. For each known customer, return the highest-gross order and separately the latest order. Use deterministic tie-breakers and explain why "highest" and "latest" are different requirements.
6. Set/absence logic. Compare UNION ALL with UNION for customer IDs coming from paid orders plus unknown-payment orders. Find customers with no orders using null-safe absence logic.
7. Date boundary. Return orders on 2026-12-10 using an inclusive-start/exclusive-end predicate. Explain why this pattern remains safe when a column later becomes a timestamp.

8. PostgreSQL thinking. In a disposable table, prove a key/quantity contract with one rejected write; demonstrate an explicit transaction followed by ROLLBACK; and save one EXPLAIN or EXPLAIN ANALYZE observation without inventing a performance claim from a tiny table.

Evidence bundle

Save G02-B-attempt.sql, observed outputs/errors, a short validation note and your assistance record under G02-B-attempt-01/. The pass rule is independent correctness across the dimensions above with no unresolved grain/cardinality/null/reconciliation/database-safety defect. Successful execution alone is not a pass.